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SCHOOL OF INDUSTRIAL AND INFORMATION ENGINEERING

MASTER OF SCIENCE IN ELECTRICAL ENGINEERING

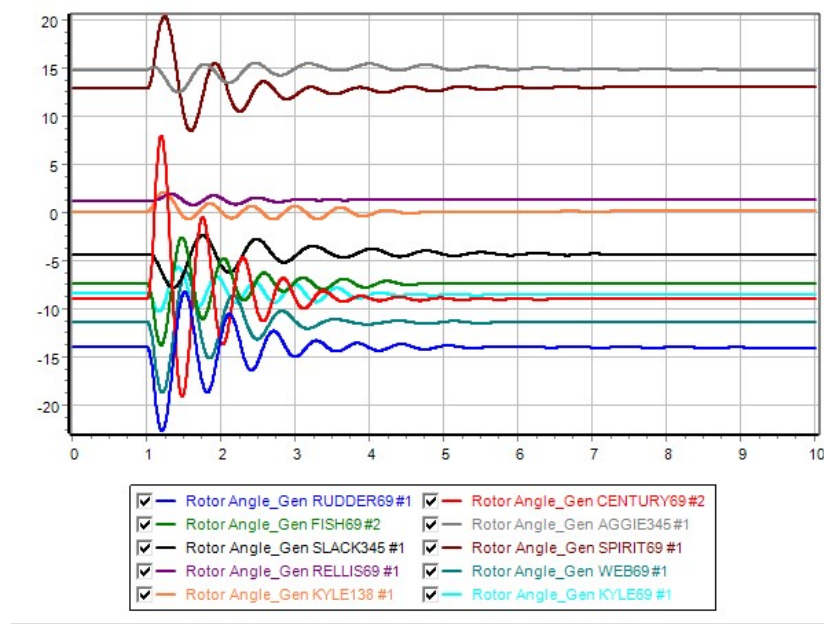
***Electric Power Systems Project 2023-2024:
Transient stability and power systems dynamics***

Academic Year 2023 - 2024

STEP 1-2

According to the guidance in step 1, we performed a transient stability analysis and obtained the following graph (*Graph 1*). The graph shows that after applying a balanced three-phase fault clearance at 1.1s, the rotor angles of all the rotors angles of generators returned to a stable state.

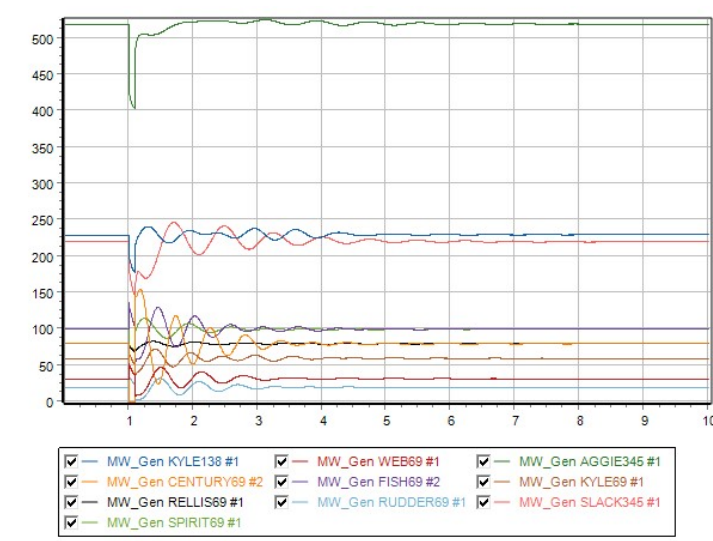
Furthermore, the rotor's angle **CENTURY69** exhibits the largest oscillation amplitude among all the generators, and the rotor's angle **RUDDER69** has the longest clearing time, taking almost 9 seconds to reach a fully stable state without any oscillation.



Fault clearance at 1.1 [s] implementation
Graph 1

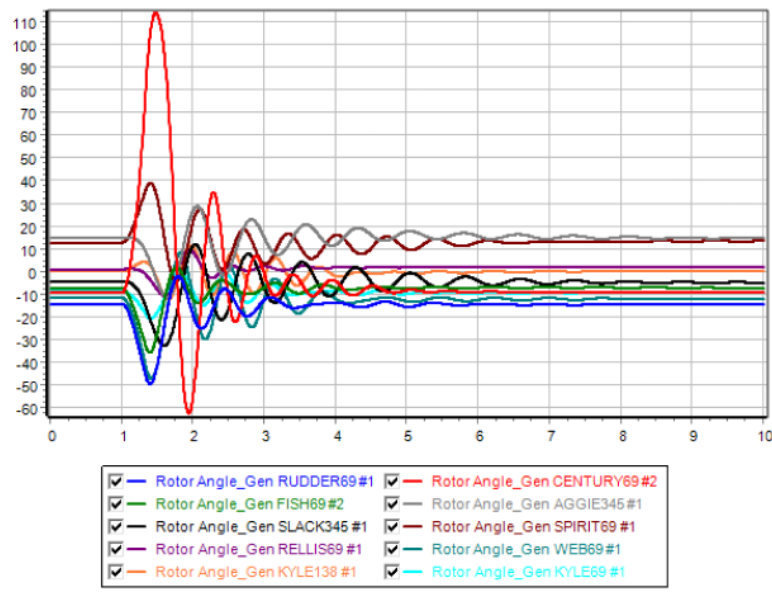
Regarding the generator power oscillation in Graph 2, Based on the power oscillation and highest amplitude of the generator KYLE138 in comparison to WEB69 it can be concluded that KYLE138 is most effected by the fault on CENTURY69.

This result is consistent with our expectation, since the higher power generation of KYLE138 leads to an increase in the critical clearing time (t_{cr}).

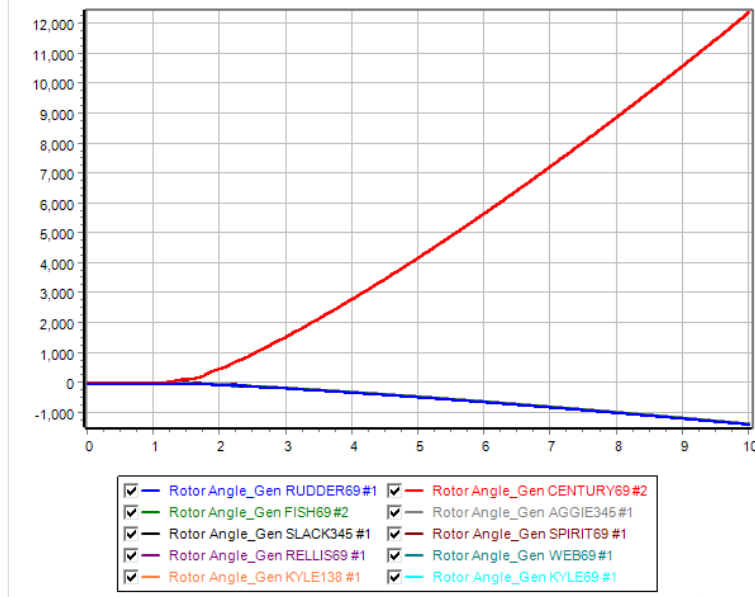


Graph 2

STEP 3



Fault clearance at 1.34 [s] implementation
Graph 3



Fault state at 1.35 [s] implementation
Graph 4

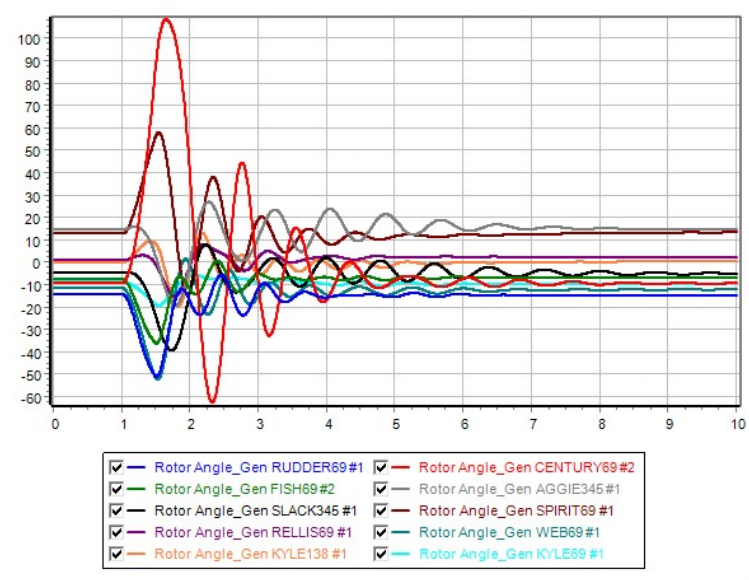
By implementing the trial-and-error method and increment of clearing time from 1.1 [s] (Table 1) by the accuracy of 0.01 [s] we find the maximum critical clearing time (0.34 [s]) which makes the oscillations vanish.

tcr	Ocillation State	Highest Amplitude
1.1	Stable	8.1
1.15	Stable	20.67
1.2	Stable	36.725
1.25	Stable	56
1.3	Stable	82.7
1.31	Stable	88.98
1.32	Stable	92.91
1.33	Stable	103.89
1.34	Stable	114.08
1.35	Unstable	Almost ∞

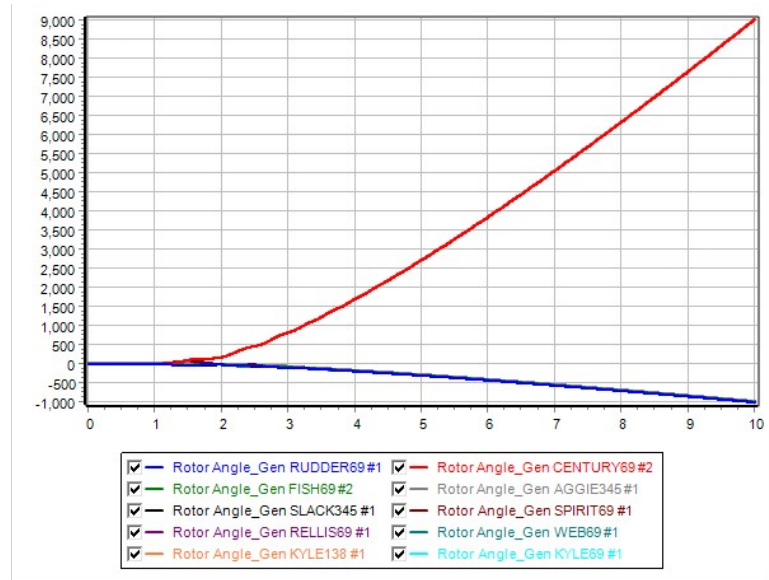
*Due to the large number of steps, some of them have been omitted
Table 1

STEP 4

We increased the inertia (H) from 3 to 5 seconds and again performed a trial-and-error method to find the maximum critical clearing time that keeps the system stable. By increasing the clearing time from 1.1s with a precision of 0.01 [s], we found that the system can withstand a fault up to 1.49 [s] without losing stability (Table 2). Graph 4 and Graph 5 show the rotor angle oscillations after the fault clearance at 1.48 [s] and 1.49 [s].



Fault clearance at 1.48 [s] implementation
Graph 5



Fault state at 1.49 [s] implementation
Graph 6

According to the theoretical analysis, increasing the inertia of the system should result in a higher critical clearing time, as shown by the following formula:

$$t_{cr} = \sqrt{\frac{4H(\delta_{cr} - \delta_o)}{w_s P_E}} \quad \text{and also} \quad t_{cr} = \frac{H}{\pi f_0 P_{max}}$$

The comparison between *Table 1* and *Table 2*, which show the results for the system with inertia 3 and inertia 5 respectively, confirms the theoretical relationship between inertia and critical clearing time. The higher the inertia, the longer the system can sustain a fault without losing stability.

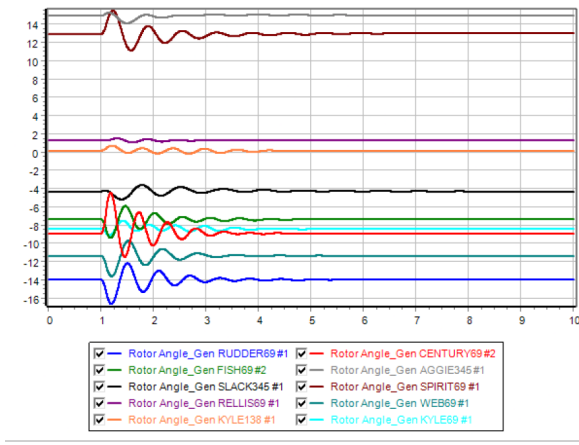
tcr	Ocillation State	Highest Amplitude
1.1	Stable	2.41
1.15	Stable	10.02
1.2	Stable	19.08
1.25	Stable	29.7
1.3	Stable	41.97
1.35	Stable	56.03
1.4	Stable	72.26
1.45	Stable	91.99
1.46	Stable	96.77
1.47	Stable	102.14
1.48	Stable	108.67
1.49	Unstable	Almost ∞

*Due to the large number of steps, some of them have been omitted

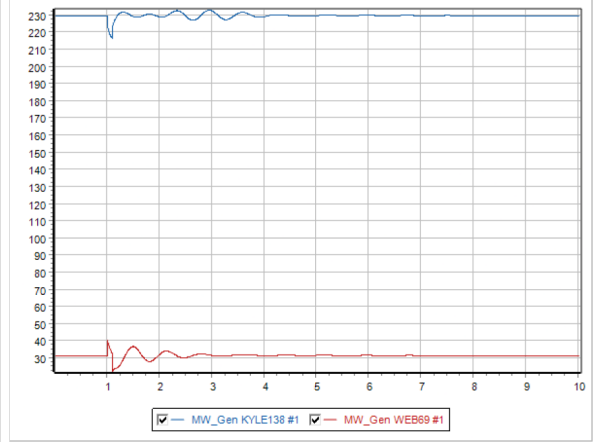
Table 2

STEP 5

We analyzed the single-line-to-ground (SLG) fault of **CENTURY69** with an inertia value of 3 [s]. *Graph 7* and *Graph 8* show the rotor's angle oscillations and generator power oscillation after the fault clearance respectively. Compared to the balanced three-phase fault (*Graph 1* and *Graph 2*), the SLG fault has a more balanced oscillation pattern, with smaller amplitude and frequency variations. This indicates that the SLG fault has a lower impact on the system stability than the balanced three-phase fault.

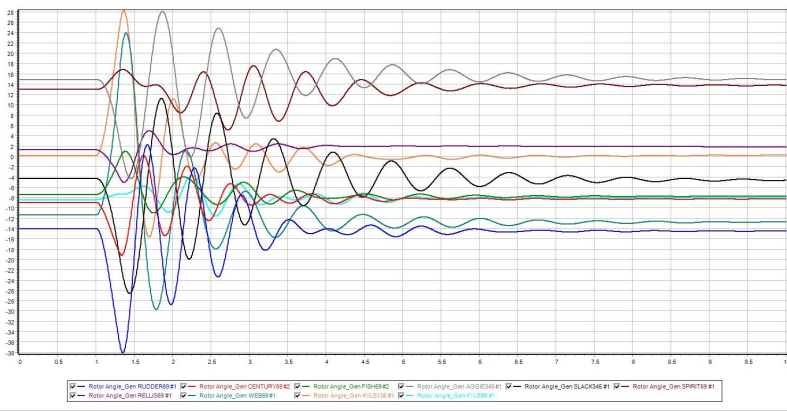


Graph 7

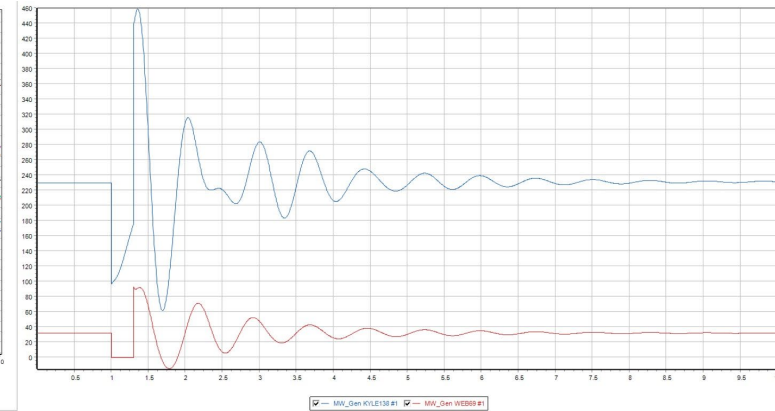


Graph 8

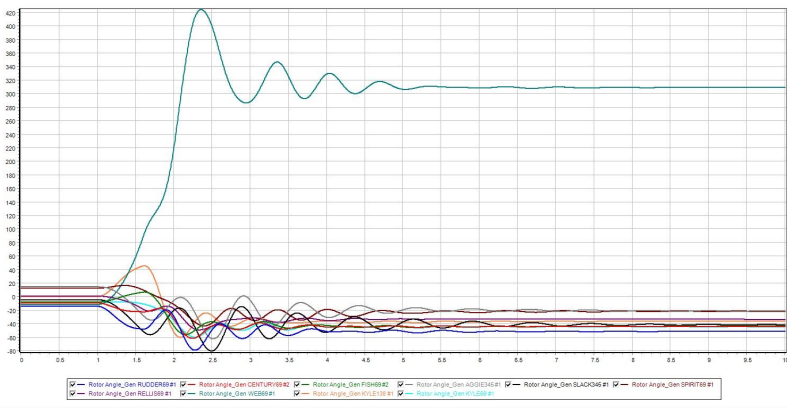
STEP 6



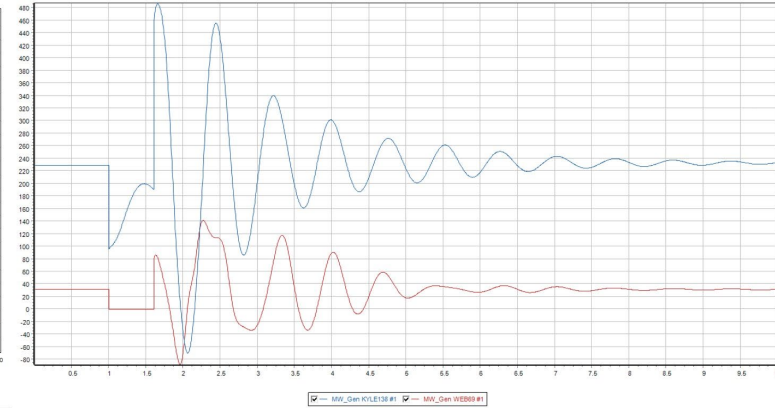
Fault clearance at 1.3 [s] implementation $P_G=32\text{MW}$
Graph 9



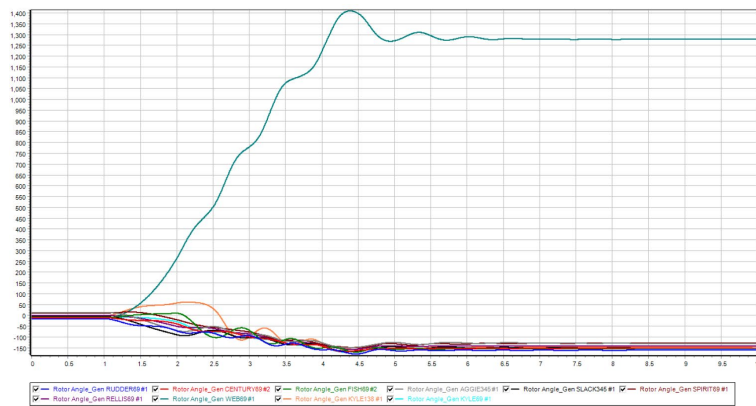
Generator power oscillation at 1.3[s]
Graph 10



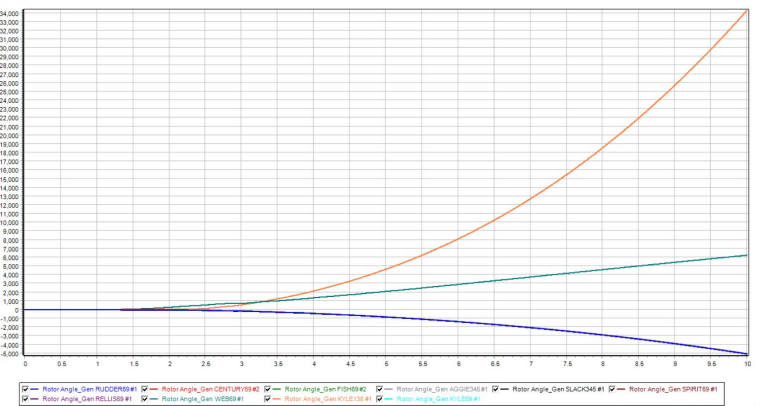
Fault clearance at 1.6 [s] implementation $P_G=32\text{MW}$
Graph 11



Generator power oscillation at 1.6[s]
Graph 12

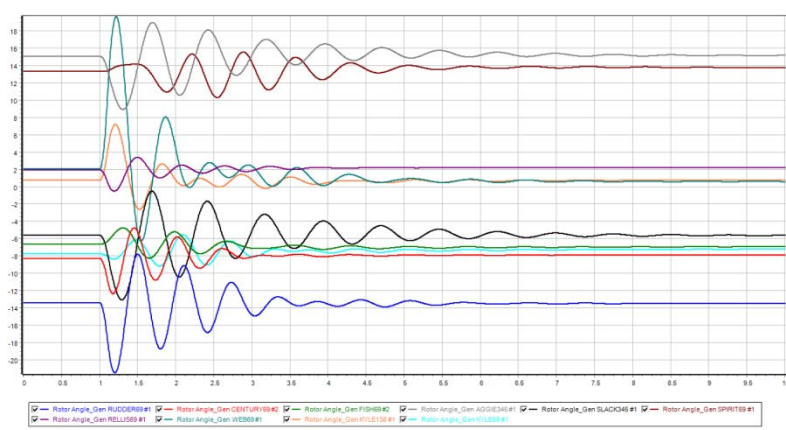


Fault clearance at 1.97 [s] implementation $P_G=32\text{MW}$
Graph 13

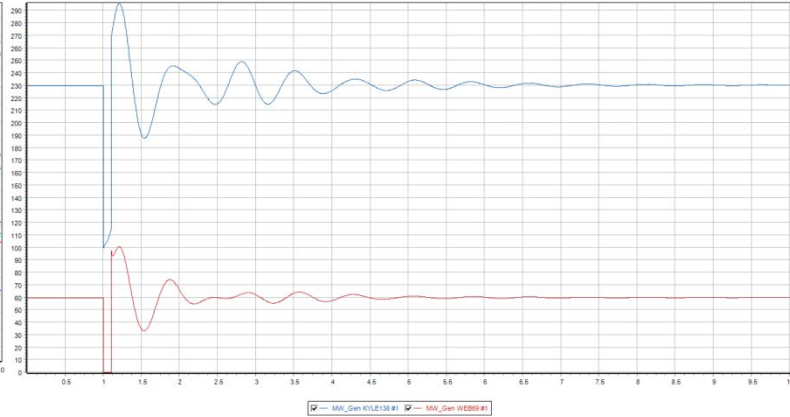


Fault state at 1.98 implementation $P_G=32\text{MW}$
Graph 14

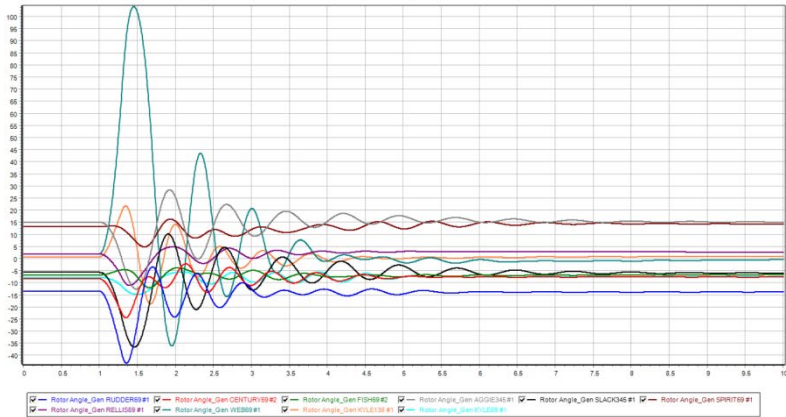
We performed the same procedure as in the previous steps to find the maximum critical clearing time of **Web69** at 32MW, before increasing its active power to 60MW. *Graph 9* and *graph 11* show the rotor angle oscillations of the generators after the fault clearance for different clearing times. We observed that the rotor angles of all the generators returned to a stable state for clearing times ranging from 1.3 [s] to 1.58 [s]. However, for clearing times from 1.59 [s] to 1.97 [s], the rotor angle of **KYLE69** reached a new equilibrium point (*graph 11* and *graph 13*), while the other generators remained stable. This indicates that **KYLE69** was more affected by the fault than the other generators. For clearing times of 1.98 [s] or higher, the system lost synchronism and became unstable (*graph 14*). Therefore, we determined that the maximum critical clearing time of **Web69** at 32MW is 0.97 [s].



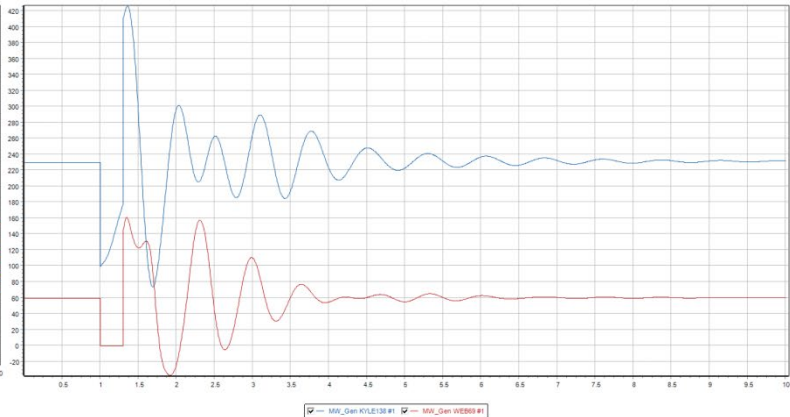
Fault clearance at 1.1 [s] implementation $P_G=60\text{MW}$
Graph 15



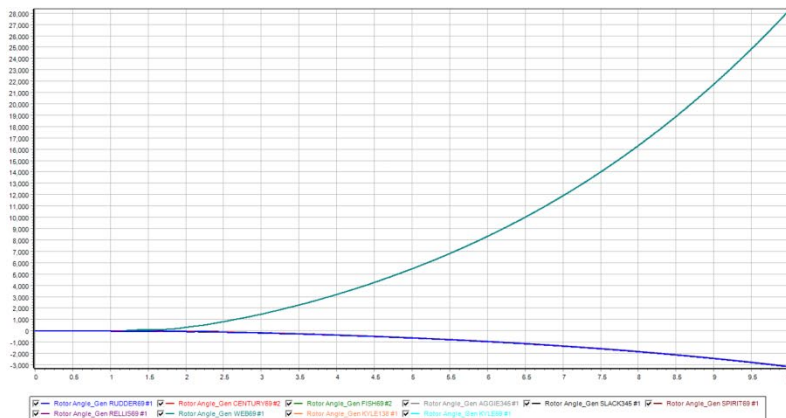
Generator power oscillation at 1.1 [s]
Graph 16



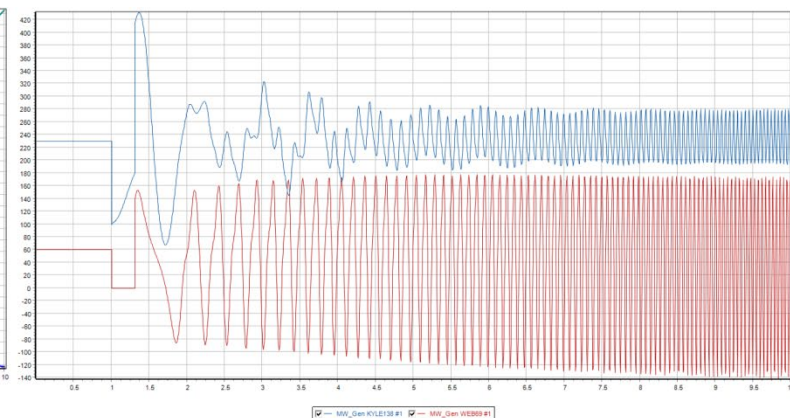
Fault clearance at 1.3 [s] implementation $P_G=60\text{MW}$
Graph 17



Generator power oscillation at 1.3 [s]
Graph 18



Fault state at 1.31 [s] implementation $P_G=60\text{MW}$
Graph 19



Generator power oscillation at 1.31 [s]
Graph 20

We increased the active power of generator **Web69** to 60MW and analyzed the impact of the fault on the system stability. *Graph 15* and *Graph 17* show the rotor angle oscillations of the generators after the fault clearance for different values of the clearing time. We noticed that the rotor angle amplitude of **KYLE69** increased significantly compared to the previous cases, and it did not reach a stable state at any new equilibrium point. Just like before the increment, this implies that **KYLE69** was highly affected by the fault and the power increase of **Web69**. The system lost synchronism and became unstable for critical time of 1.31 [s] or higher (*Graph 19*). Therefore, we determined that the maximum critical clearing time of **Web69** at 60MW was 0.30 [s]. This result confirms that increasing the generator power reduces the critical clearing time, as expected from the theoretical analysis. The higher the power generation of Web69, the larger the **area1** and the shorter the clearing time. (Due to larger system is more stable)

STEP 7

In the equivalent area criterion (EAC) method, we consider the worst-case scenario, which assumes that the output power of the faulted area (P_E) is zero during the fault. However, in a realistic situation, P_E will not be zero, but rather a fraction of its pre-fault value. This means that the power imbalance between the two areas ($P_n - P_E - P_L$) is smaller than in the worst-case scenario, and therefore the accelerating torque on the faulted area is also smaller. This results in a lower rotor angle deviation and a higher critical clearing time (t_{cr}) for the fault. Thus, the EAC method provides a conservative estimate of the system stability, and the actual system may be able to withstand longer faults than predicted by the EAC method. These facts explain the difference between the values of the critical clearing time (t_{cr}) obtained by the two methods. (*Table 3*)

contingency type	Total time(s)		Delta time (s)	
	PW	Matlab(EAC)	PW	Matlab(EAC)
Century69	1.34	1.2921	0.34	0.2921
Fish69	1.29	1.2603	0.29	0.2603
Kyle138	1.27	1.2428	0.27	0.2428
Sprit69	1.17	1.1445	0.17	0.1445
Web69	1.58	1.3927	0.58	0.3927
Web69	1.98	—	0.98	—
	New equilibrium point			

The critical clearing time getting from PW and MATLAB (EAC)

Table 3